Mini-Test

Modern Physics

Gergő Dénes,[1,*] Réka Szilvási,[2,†] Angelo Valli,[1,‡] and Dániel Varjas[1,§]

[1] *Department of Theoretical Physics, Institute of Physics,*
Budapest University of Technology and Economics, Műegyetem rkp. 3., H-1111 Budapest, Hungary
[2] *Department of Nuclear Techniques, Institute of Nuclear Techniques,*
Budapest University of Technology and Economics, Műegyetem rkp. 3., H-1111 Budapest, Hungary

(Dated: April 20, 2026)

Text and solution for Monday's mini-test.

* denes.gergo@edu.bme.hu
† szilvasi.reka@ttk.bme.hu
‡ valli.angelo@ttk.bme.hu
§ varjar.daniel@ttk.bme.hu

WEEK VIII

- A hydrogen atom is in an excited $4d/5f$ state in *zero external magnetic field* ($B = 0$).
  - What is the energy of such a state? (0.5pt)
  - What is the number (g) of $4d/5f$ states with equal energy? (0.5pt)

When the atom is placed in a magnetic field $B = 5\,\mathrm{T}$, the $4d/5f$ subshell is split by the potential $V_B = -\mu_B m B$ (*Zeeman effect*).

  - Calculate the energy shift ΔE of the level with *maximal/minimal* m. (1pt)

Ignore the electron spin. d *orbitals have* $\ell = 2$ *and* f *orbitals have* $\ell = 3$. *Use:* $\mu_B = 5.788 \times 10^{-5}\,\mathrm{eV/T}$. *Give energies in* eV *units.*

Solution
In zero magnetic field, the energy of the shell depends *only* on the principal quantum number n

$$E = -\frac{1}{n^2}E_0$$

with $E_0 = 13.6\,\mathrm{eV}$. Therefore

$$E_4 = -E_0/16 = -0.85\,\mathrm{eV} \qquad \text{and} \qquad E_5 = -E_0/25 = -0.544\,\mathrm{eV}.$$

Ignoring the intrinsic spin of the electron, the *degeneracy* g is controlled by the quantum numbers ℓ and $m = \{-\ell, \ldots, -1, 0, 1, \ldots, \ell\}$, therefore the number of states in each subshell is $g = 2\ell + 1$. For the d shell $g = 5$, and for the f shell $g = 7$.

The energy shift is calculated according to the potential $\Delta E = V_B$, for $m = \pm\ell$ for the *maximum* and *minimum* cases, respectively. Therefore, for the d shell

$$\Delta E_{\mathrm{d,max}} = -2\mu_B B = -5.79 \times 10^{-4}\,\mathrm{eV} \qquad \text{and} \qquad \Delta E_{\mathrm{d,min}} = 2\mu_B B = 5.79 \times 10^{-4}\,\mathrm{eV}$$

and for the f shell

$$\Delta E_{\mathrm{f,max}} = -3\mu_B B = -8.68 \times 10^{-4}\,\mathrm{eV} \qquad \text{and} \qquad \Delta E_{\mathrm{f,min}} = 3\mu_B B = 8.68 \times 10^{-4}\,\mathrm{eV}$$

Note that the potential has a negative sign, which means that it energetically favors the anugular momentum to be parallel to the magnetic field *aligned*, so that the shift is *negative* for the state with $m > 0$ (corresponding to the case in which $\vec{L}$ is parallel to $\vec{B}$) and *positive* for the state with $m < 0$ ($\vec{L}$ is anti-parallel to $\vec{B}$).